BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022 • Paper Code: 8193

Retyped faithfully from the original board paper — same questions, no changes.

Mathematics - Paper: II - Objective (ii)

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 8193

Note: Note: - You have four choices for each objective type question as A, B, C and D. The choice which you think is correct; fill that circle in front of that question number in your answer book. Use marker or pen to fill the circles. Cutting or filling up two or more circles will result no mark.

Note: SECTION-A

Q1. Length of latus-rectum of ellipse $x^2/a^2 + y^2/b^2 = 1$, is:

(A) $2a^2/b$ (B) $2b^2/a$

(C) $b^2/2a$ (D) $b/2a^2$

Q2. Centre of the Hyperbola $(x+1)^2/9 - (y-2)^2/4 = 1$, is:

(A) (1,2) (B) (2,1)

(C) (-1,2) (D) (1,-2)

Q3. Unit vector perpendicular to a and b is:

(A) $(a\times b)/(|a||b|)$ (B) $(a.b)/(|a||b|)$

(C) $(a.b)/(a\times b)$ (D) $(a\times b)/(|a\times b|)$

Q4. $2i \cdot j \times k =$

(A) 2 (B) Zero

(C) 1/2 (D) ∞

Q5. $\text{Sinh}^{-1}x =$

(A) $(e^x+e^{-x})/2$ (B) $(e^x-e^{-x})/2$

(C) $(e^x-e^{-x})/(e^x+e^{-x})$ (D) $(e^x+e^{-x})/(e^x-e^{-x})$

Q6. If $g(x) = 1/x^3$, $x \neq 0$, then $g \circ g(x) =$

(A) x^2 (B) $1/x^2$

(C) x^6 (D) $1/x^4$

Q7. Derivative of $(x^3+1)^9$ w.r.t. x^3 equals:

(A) $9(x^3+1)^8$ (B) $27x^2(x^3+1)^8$

(C) $3x(x^3+1)^8$ (D) $27(x^3+1)^8$

Q8. If $\lim(x \to a) (f(x)-f(a))/(x-a)$ exists, then it is equal to:

(A) f'(x) (B) f'(a)

(C) zero (D) ∞

Q9. The distance of the line $4x - 3y - 25 = 0$ from the origin is:

(A) 1 (B) 5

(C) 25 (D) 2

Q10. Normal form of the equation of straight line is:

(A) $y = mx + c$ (B) $y - y_1 = m(x - x_1)$

(C) $x/a + y/b = 1$ (D) $x\cos\alpha + y\sin\alpha = p$

Q11. $2x + y < 6$ is satisfied by which point?

(A) (3,1) (B) (1,3)

(C) (0,7) (D) (4,0)

Q12. Equation of the tangent to the circle $x^2 + y^2 = 4$ at (1,3) is:

(A) $x + 3y = 4$ (B) $x - 3y = 4$

(C) $3x + y = 4$ (D) $3x - y = 4$

Q13. $\int 2^x dx =$

(A) $2^x + c$ (B) $2^x.\ln 2 + c$

(C) $(\ln 2/2^x) + c$ (D) $(1/\ln 2).2^x + c$

Q14. $\int_0^3 1/(9+x^2)\, dx =$

(A) $\pi/4$ (B) $\pi/6$

(C) $\pi/12$ (D) $\pi/8$

Q15. Slope of vertical line is:

(A) 0 (B) ∞

(C) 1 (D) -1

Q16. Two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$, are perpendicular if:

(A) $a_1a_2 + b_1b_2 = 0$ (B) $a_1a_2 - b_1b_2 = 0$

(C) $a_1b_2 + a_2b_1 = 0$ (D) $a_1b_2 - a_2b_1 = 0$

Q17. If $y = \ln(x^2)$, then $dy/dx =$

(A) $1/x^2$ (B) $1/2x^2$

(C) $2/x$ (D) $2/x^2$

Q18. $d/dx(\cos x^2) =$

(A) $-\sin x^2$ (B) $-2x \sin x^2$

(C) $2x \sin x^2$ (D) $-x \sin x^2$

Q19. $\int e\hat{}(\tan^{-1}x) \cdot 1/(1+x^2)\, dx =$

(A) $e\hat{}(\tan^{-1}x) + c$ (B) $1/(1+x^2) + c$

(C) $e\hat{}(\cos^{-1}x) + c$ (D) $e\hat{}(\sec^{-1}x) + c$

Q20. $\int e^x(1+x)dx =$

(A) $e^x + c$ (B) $xe^x + c$

(C) $x^2 + c$ (D) $1/2\, x^2e^x + c$

Mathematics - Paper: II - Subjective

Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Note: Note: - Section B is compulsory. Attempt any three questions from section C.

Note: SECTION-B

Q2.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Determine whether the function f(x) = sin x + cos x is even or odd.

ii. Find the composition function fof(x) if $f(x) = 1/\sqrt{(x-1)}$

iii. Express $\lim(n\to\infty)(1 + 3/n)^{2n}$ in term of "e".

iv. What is the implicit function?

v. Evaluate $\lim(x\to0)\ \sin x°/x$

vi. Differentiate (2x-3)/(2x+1) w.r.t. x

vii. Find dy/dx if $y^2 - xy - x^2 + 4 = 0$

viii. Find derivative of $(\sin2\theta - \cos3\theta)^2$ w.r.t θ

ix. If $y = x^2\ln(1/x)$, find dy/dx

x. Find y_2 if $y = x^2e^{-x}$

xi. Find the intervals in which $f(x) = 4 - x^2$; $x \in (-2,2)$ is increasing or decreasing.

xii. Differentiate $\log_{10}(ax^2 + bx + c)$ w.r.t. x

Q3.. (Marks 16) Write short answers to any Eight parts. (8 x 2 = 16)

i. Using differentials find dy/dx in the equation $xy + x = 4$

ii. Evaluate $\int \sin^2 x \, dx$

iii. Evaluate $\int \sec^2 x/\sqrt{\tan x} \, dx$

iv. Evaluate $\int \tan^{-1} x \, dx$

v. Evaluate $\int e^{-x}(\cos x - \sin x) \, dx$

vi. Evaluate $\int (3x+1)/(x^2-x+6) \, dx$

vii. Evaluate $\int_0^2 \ln x \, dx$

viii. Find the area between the x-axis and the curve $y = 4x - x^2$

ix. Show that the points A(0,2), B(√3,-1) and C(0,-2) are vertices of a right triangle.

x. Find an equation of line passing through A(-5,-3) and B(9,-1).

xi. Find an equation of the line through (-4,7) and parallel to the line $2x - 7y + 4 = 0$

xii. Find lines represented by $3x^2 + 7xy + 2y^2 = 0$

Q4.. (Marks 18) Write short answers to any Nine parts. (9 x 2 = 18)

i. How would you obtain the optimal solution from feasible region?

ii. Indicate the solution region of $3x - 2y \geq 6$

iii. Find an equation of the circle passing through A(1,4), B(-1,8) and tangent to the line $x + 3y - 3 = 0$

iv. Write down the equation of normal to circle $3x^2 + 3y^2 + 5x - 13y + 2 = 0$ at (1, 10/3).

v. Investigate vertex and directrix of $x + 8 - y^2 + 2y = 0$

vi. Form the equation of ellipse from that data, foci (±3,0) and minor axis of length 10.

vii. Find foci and eccentricity of $y^2/16 - x^2/9 = 1$

viii. Use vectors, to prove that the diagonals of a parallelogram bisect each other.

ix. For what values of a and b from the given parallel vectors $3i - j + 4k$ and $ai + bj - 2k$

x. Prove that in any triangle ABC: $a = b \cos C + c \cos B$

xi. Find area of parallelogram, where vertices are A(-1,1,1), B(-1,2,2), C(-3,4,-5) and D(-3,5,-4).

xii. Find α from the given coplanar vectors i - 2αj - k, i - j + 2k and αi - j + k

xiii. A force F = 3i + 2j - 4k is applied at the point (1,-1,2). Find the moment of the force about the point (2,-1,3)

> Note: SECTION-C (Each question carries 10 marks)

Q5.. (Marks 5) (a) If f(x) = {3x if $x \leq -2$; x^2-1 if $-2<x<2$; 3 if $x \geq 2$} Discuss the continuity at x = 2 and x = -2 (b) Differentiate $(x^2+1)/(x^2-1)$ w.r.t $(x-1)/(x+1)$

Q6.. (Marks 5) (a) Evaluate $\int dx/\sqrt{(7 - 6x - x^2)}$ (b) Find h such that the points A(h,1), B(2,7) and C(-6,-7) are vertices of a right triangle.

Q7.. (Marks 5) (a) Find area between the x-axis and the curve $y = \sqrt{(2ax - x^2)}$ when $a > 0$ (b) Graph the feasible region of the following system of linear inequality and find corner points $2x+3y\leq18$, $2x+y\leq10$, $x+4y\leq12$, $x\geq0$, $y\geq0$

Q8.. (Marks 5) (a) Find a joint equation of the straight lines through the origin and perpendicular to the lines represented by $x^2 + xy - 6y^2 = 0$ (b) Find an equation of the parabola whose focus is F(-3,4) and directrix is 3x - 4y + 5 = 0

Q9.. (Marks 5) (a) Find centre, foci, eccentricity and vertices of hyperbola $4x^2 - 8x - y^2 - 2y - 1 = 0$ (b) Prove that $\sin(\alpha - \beta) = \sin\alpha \cos\beta + \cos\alpha \sin\beta$